

GLC

THE GENERAL LAW OF COGNITION

Derived, not postulated.

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Opening — The Question

Every theory of mind has to answer the same question: what are the necessary and sufficient structural conditions for a cognitive system to produce output at all? Not what the output contains. Not what the system is made of. What structural conditions must simultaneously hold for anything recognizable as cognition to occur?

GLC’s answer: three conditions. Generative motion. Structural constraint. Verification. All three simultaneously. The multiplicative structure is not a modeling choice — it follows from what happens when any one of the three is absent. And those three conditions are not postulated. They are derived from seven invariants that established science has confirmed in every known self-organizing system. This document shows that derivation precisely.

Section 1 — The Seven Invariants

Before deriving GLC, establish what the Seven Invariants are and why a cognitive system is bound by them. This is the section that replaces the negative argument (‘why would cognition be exempt?’) with the positive case.

1.1 — What the Invariants Are

The Seven Invariants are not cognitive physics claims. They are conditions observed in every known self-organizing system across independent scientific fields. They were established independently, without reference to cognition, by researchers studying thermodynamics, dynamical systems, information theory, systems biology, dissipative structures, mathematical physics, and complex systems theory.

Invariant	Definition and Source
1 · Conservation	Thermodynamics: energy neither created nor destroyed. Information theory: Shannon entropy conserved across lossless transforms. Outputs cannot exceed inputs. No perpetual motion of any kind.Clausius (1850, 1865) · Shannon (1948)
2 · Continuity	Dynamical systems: physical state evolves without discontinuous jumps. All change traceable through continuous evolution. No teleportation between states.Strogatz (1994)
3 · Quantization	Information theory: a bit is 0 or 1, not 0.73. Stable states are discrete. No infinitely fine-grained state is stable or storable.Shannon (1948)

4 · Coherence	Systems biology: contradictory metabolic pathways don't produce compromise — they produce failure. Contradictory states cannot coexist in a functioning system.Kauffman (1993)
5 · Entropy/Attractor	Prigogine's dissipative structures: far-from-equilibrium systems produce local entropy while maintaining global organization through attractor dynamics. Motion converges, not expands forever.Prigogine & Stengers (1984) · Prigogine Nobel (1977)
6 · Symmetry Protection	Noether's theorem: every conservation law corresponds to a symmetry. Verified states resist arbitrary change. What is conserved is protected.Noether (1918)
7 · Hierarchy	Complex systems theory: stable complexity builds from verified stable layers. Each layer depends on verified completion of prior layers. No structure can rest on unverified foundations.Simon (1962)

1.2 — Why Cognition is Bound: The Positive Argument

A cognitive system is a self-organizing information-processing system. The Seven Invariants are not properties of specific substrates — they are properties of stable self-organized structure itself. They operate at the level of the system's organization, not its physical implementation. The evidence is empirical and direct. When any of the seven conditions is violated in a cognitive system, the specific failure mode that condition predicts appears. This is not substrate reduction — it is observation at the cognitive level. The predicted violations match the observed failures:

Invariant Violated	Cognitive Failure Observed in Deployed AI Systems
Invariant 1 violated	Hallucination. Outputs produced with no traceable connection to verified inputs. The system cannot distinguish knowledge from noise.
Invariant 2 violated	Identity drift. Reasoning disconnected between sessions or turns. Each state appears in isolation rather than as part of a continuous trajectory.
Invariant 3 violated	Perpetual hedging. No discrete knowledge state exists. The system produces continuous confidence values but can never commit to knowing.
Invariant 4 violated	Silent contradiction. Conflicting beliefs coexist without resolution. Confident outputs from both sides of a contradiction at different turns.
Invariant 5 violated	Infinite regress. Novelty generates novelty. Exploration expands without attractor-driven convergence. No settled knowledge is produced.
Invariant 6 violated	Invisible rewriting. Prior verified conclusions silently revised without acknowledgment, new evidence, or logged change.
Invariant 7 violated	Floating conclusions. Reasoning chains built on unverified premises. Local logic appears coherent; global structure has no verified foundation.

These failures are documented across the literature (Huang et al., 2023). The match between predicted violations and observed failures is the empirical evidence that the invariants bind at the cognitive level. The question is not ‘why would cognition be exempt?’ The question is: given that these violations produce exactly these failures, what would exemption even mean?

Section 2 — From Seven Invariants to Three Terms

This is the derivation. Not an assertion with citations — a step-by-step proof that the Seven Invariants, applied simultaneously to a cognitive system, partition into exactly three functional clusters. Those clusters are M, S, and V.

2.1 — The Partitioning Argument

Start with a precise question: given that all seven invariants must hold simultaneously in a cognitive system, what functional roles must that system have to satisfy all of them? The answer requires examining what each invariant demands of the system’s behavior — not what each invariant is, but what it forces the system to do.

CLUSTER 1 — INVARIANTS 2 + 5 → M (Generative Motion)

Invariant 2 — Continuity demands:

The system must be in continuous motion. State cannot be static — it must evolve. This demands a generative function: something that keeps the system moving through hypothesis space. Without it, $d\Psi/dt = 0$ and the system is inert. An inert cognitive system is not a cognitive system at all.

Invariant 5 — Attractor demands:

That motion must be directed — it cannot expand indefinitely. The generative function must produce motion that converges toward attractor states. Without Invariant 5, continuous motion (Invariant 2) would produce infinite regress: novelty generating novelty with no convergence. The system would move forever without producing anything.

⇒ **Together, Invariants 2 and 5 require: continuous directed motion through hypothesis space.**
This is M. M is not optional. It is what Invariants 2 and 5 force the system to have.

CLUSTER 2 — INVARIANTS 4 + 7 → S (Structural Constraint)

Invariant 4 — Coherence demands:

The system cannot hold contradictory states. Every new prediction must be checked against existing state for consistency. This demands a constraint function: something that organizes the prediction space so that new motion doesn’t produce contradiction. Without it, M generates novelty that contradicts prior verified state — silent contradiction accumulates.

Invariant 7 — Hierarchy demands:

Each new prediction must rest on verified foundations. Stable complexity builds from verified stable layers. Without Invariant 7, M generates predictions that float above unverified premises — the structure looks coherent locally but has no verified foundation globally. This is the formal mechanism of floating conclusions.

⇒ **Together, Invariants 4 and 7 require: organized verified structure that constrains M’s motion.**

This is S. S is not optional. It is what Invariants 4 and 7 force the system to have.

CLUSTER 3 — INVARIANTS 1 + 3 + 6 → V (Verification Gate)

Invariant 1 — Conservation demands:

Outputs must trace to verified inputs. The system cannot generate knowledge from nothing. This demands an accounting function: something that traces each output back to its verified source. Without it, the system cannot distinguish knowledge from confabulation.

Invariant 3 — Quantization demands:

Knowledge states are discrete. There must be a gate that determines which state is active — prediction or knowledge. This is not a gradient. A system in prediction state with confidence 0.99 is not in knowledge state. The transition is discrete. Without a gate, the system permanently inhabits prediction state with no mechanism to exit.

Invariant 6 — Symmetry Protection demands:

Verified states resist arbitrary change. Once a conclusion is verified and sealed, it must be protected from silent revision. Only new verified evidence can change it. Without this, every prior conclusion is vulnerable to retroactive revision — the system cannot accumulate stable knowledge across time.

⇒ Together, Invariants 1, 3, and 6 require: a verification gate that accounts, transitions, and seals.

This is V. V is not optional. It is what Invariants 1, 3, and 6 force the system to have.

2.2 — Why Exactly Three Clusters

The partition into M, S, V must be shown to be both exhaustive and non-redundant. If it is not exhaustive, some invariant is unaccounted for. If it is not non-redundant, a simpler partition would suffice.

Question	Answer
Exhaustive?	Every invariant is assigned to exactly one cluster by its primary functional demand. Invariant 2 demands continuous motion → M. Invariant 5 demands directed convergence → M. Invariant 4 demands contradiction prevention → S. Invariant 7 demands verified foundations → S. Invariant 1 demands output-input tracing → V. Invariant 3 demands discrete state gate → V. Invariant 6 demands sealed state protection → V. No invariant is unaddressed. The partition is exhaustive.
Non-redundant?	No invariant belongs to a different cluster by its primary function. Conservation (1) governs output-input tracing — that is verification, not motion or structure. Continuity (2) governs state evolution — that is motion, not structure or verification. The assignments follow from what each invariant demands, not from a choice of how to group them.
Can fewer than 3 work?	No. Two clusters cannot simultaneously satisfy both the generative requirement (Invariants 2+5 demand continuous directed motion) and the constraint requirement (Invariants 4+7 demand organized verified structure) without collapsing into a system that either moves without constraint or constrains without motion. And the epistemic gate (Invariants 1+3+6) cannot be derived from either of the other two clusters — this is proven by the Epistemic Separation Theorem (T365). Minimum is three.

Can more than 3 work?	Additional components are always implementations of M, S, or V, not new functional categories. The three clusters are the complete minimal set. Any cognitive system with more apparent components either has redundant components (multiple implementations of one cluster) or can be reduced to these three functional roles.
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2.3 — Why Multiplication

Addition would imply: some M can compensate for missing S, or some V can compensate for missing M. That would mean partial cognitive output is possible with any single term present. But the invariants are all binding simultaneously — you cannot satisfy Invariant 2 (continuous motion) while violating Invariant 4 (coherence) and still have a functioning cognitive system. Each invariant cluster is necessary, not compensatory.

The precise argument: if any one cluster is at zero, at least one invariant in that cluster is being violated. A violated invariant produces the corresponding failure mode. Those failure modes are not ‘reduced output’ — they are specific system failures (hallucination, drift, perpetual hedging). The system does not produce ‘some’ knowledge with $V = 0$. It produces prediction labeled as knowledge indefinitely. Zero in any term = system failure, not diminishment. Multiplication encodes this necessity. Addition does not.

2.4 — The Result

DERIVATION RESULT
Seven invariants, applied simultaneously to a cognitive system,
partition into exactly three necessary functional clusters:
Invariants 2 + 5 → continuous directed motion → M
Invariants 4 + 7 → organized verified structure → S
Invariants 1+3+6 → verification gate that seals → V
All three required simultaneously.
Zero in any one = at least one invariant violated = specific failure mode.
C = M × S × V

This is not a postulate. It is the consequence of all seven invariants

holding simultaneously in a self-organizing cognitive system.

Section 3 — The Canonical Form

With the derivation complete, the law can be stated cleanly.

$$C = M \times S \times V$$

C = cognitive output (knowledge production rate)

M = generative momentum

continuous directed motion through hypothesis space

engine of the exploration phase

derives from Invariants 2 + 5

S = structural constraint density

organized verified structure

gives M something to work against productively

determines how fast V can operate

derives from Invariants 4 + 7

V = verification

the epistemic gate

determines when prediction becomes knowledge

seals and protects verified conclusions

derives from Invariants 1 + 3 + 6

The knowledge state is stable: Knowledge = the cognitive state that remains after V brings the admissible hypothesis space to one. Not belief. Not prediction. Not extendable by M alone. Changes only when verified constraints change. Discrete and enforceable.

THE CONVERGENCE EVIDENCE

GLC was derived independently from observation of cognitive behavior. The Seven Invariant mapping was discovered afterward — two independent paths arriving at the same three-term structure. Independent convergence eliminates reverse-engineering as explanation.

CIL (Civilizational Intelligence Law) was derived independently from observing civilizational progress patterns: $\text{Progress} \approx (\text{Rate} \times \text{ConstraintDensity} \times \text{Verification}) / \text{Cost}$. Found afterward to map directly to $M \times S \times V$ with a cost normalization. Third independent derivation. Same structure. The law was always there.

Section 4 — The Phase Structure

Before the rate interpretation, establish precisely what M and V are operating on: the hypothesis space. The phase structure makes the M/V relationship exact rather than intuitive.

Phase	Definition
Exploration Phase $ H V(C) > 1$	More than one hypothesis remains admissible under current verified constraints. M is lawful in this phase. Creativity, novelty, and prediction all operate here. V has not yet closed the space. This is the pre-knowledge state: high confidence is possible, but knowledge is not yet produced.
Enforcement Phase $ H V(C) = 1$	Verified constraints have eliminated all alternatives but one. V closes to this state by applying constraints from S against the active hypothesis space. Creativity is not lawful in this phase — there is nothing left to explore. This is the knowledge state: the stable cognitive state that remains after closure.

The knowledge state transition fires when V’s action on the hypothesis space reduces it from $|H|V(C)| > 1$ to $|H|V(C)| = 1$. This is Invariant 3 (Quantization) operationalized: a discrete event, not a gradient. High confidence in the exploration phase and actual knowledge in the enforcement phase feel identical from the inside. Only V can distinguish them.

M operates in the exploration phase. V closes the exploration phase. S is what makes closing possible — verified structure provides the constraints V applies to eliminate alternatives. If S is sparse or unverified, V has little to work with. The hypothesis space stays open regardless of how hard V tries to close it.

The Phase Boundary is the Formal Statement of Why Creativity and Knowledge are Mutually Exclusive
Creativity requires $ H V(C) > 1$ (alternatives must exist to explore)
Knowledge requires $ H V(C) = 1$ (all alternatives eliminated)
You cannot be in both states simultaneously.
M and V are not concurrent competitors — they are phase-specific operations.
M opens. V closes. The transition is discrete.

Section 5 — The Rate Interpretation (GLC v2)

With the phase structure established, the rate interpretation follows naturally. M opens the hypothesis space. V closes it. The rate at which each happens — and the structural capacity that determines V’s ceiling — gives GLC its measurement-level expression.

The conceptual derivation:

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M is novelty — the prediction space opening.
V is verification — the prediction space closing.
S_v is the verified structural density that determines
how fast closing can happen.

Stripped to first principles:
M   = how fast new predictions arrive
      (novelty arrival rate)
V   = how fast verified constraints close them
      (verification throughput)
S_v = the capacity of the constraint mesh
      (determines V’s ceiling)
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GLC v2 — Rate Form (Measurement-Level Expression)	
C = M × S_v × V	
M	= dP_open/dt
Rate of hypothesis space opening	
Novelty arrival rate	
Invariants 2 + 5	
S_v	= E_t · MVR where MVR = E_v / E_t
Verified structural density (from DYL)	
E_v = verified edges (survived contradiction)	
E_t = total edges in knowledge mesh	
Capacity ceiling for V throughput	

Invariants 4 + 7
V = dP_close/dt
Verification throughput
Rate at which V closes the hypothesis space
toward $ H V(C) = 1$
Invariants 1 + 3 + 6
Key relationship: V_max = f(S_v)
High S_v → rich verified constraint mesh
→ each hypothesis has more constraints to close against
→ high V throughput ceiling
Low S_v → sparse or unverified mesh
→ few constraints available per hypothesis
→ low V ceiling regardless of V effort

Why this is the same law: $C = M \times S \times V$ is unchanged. GLC v2 specifies what M and V are measured as — rates of hypothesis space opening and closing. S becomes S_v, the verified structural density that DYL shows is the measurable form of S. Same derivation, same structure, measurement-level precision added.

The BLC connection made explicit: BLC describes the cognitive cycle as inhalation (M fires, hypothesis space opens) and exhalation (V fires, hypothesis space closes). GLC v2 gives this the rate mathematics. A healthy cognitive cycle is one where V throughput keeps pace with M arrival rate. When it doesn't, prediction debt accumulates — the formal mechanism behind hallucination and context drift over extended sessions.

Section 6 — Three Failure Modes, Three Fixes

The rate interpretation makes failure diagnosis precise. Each failure mode traces to a specific invariant cluster violation, then to a specific rate problem, then to a specific fix.

FAILURE 1 — $M \rightarrow 0$

Invariant violation: Continuity (2) + Attractor (5)

System loses directed generative motion. Hypothesis space stops opening. Observable as: confirms without discovering. High confidence, no novelty. Expert paralysis. $dP_{open}/dt \rightarrow 0$.

Fix: Adversarial load (GLC-12). Introduce novel constraints that force the exploration phase to reopen. Expand the trust region to force M.

FAILURE 2 — $S_v \rightarrow 0$ (S_{raw} may be high)**Invariant violation: Coherence (4) + Hierarchy (7)**

Structure exists but is unverified. High mesh density, low MVR. Two systems with identical S_{raw} perform differently under adversarial pressure — the one with higher MVR closes predictions; the other produces noise. V ceiling is low because few verified constraints exist to close incoming hypotheses against.

Fix: State Engineering. Build and verify the mesh before applying pressure. Run contradiction attempts on existing structure to raise E_v . S_v first, then M. Sequence matters.

FAILURE 3 — $V \rightarrow 0$ (verification bottleneck)**Invariant violation: Conservation (1) + Quantization (3) + Symmetry (6)**

Epistemic gate absent or overwhelmed. $dP_{open}/dt \gg dP_{close}/dt$. Prediction debt accumulates. Observable as: perpetual hedging, hallucination, confident wrongness. Dual form (CIL-E): invalid hypothesis elimination rate falls below generation rate $\rightarrow$ progress $\rightarrow 0$.

Fix: Rate-limit M, or enrich S_v to raise the V ceiling. Cannot fix a verification bottleneck by trying harder to verify on a sparse mesh. Fix sequence: enrich S_v first (verified constraints), then V can close faster.

The critical interaction: Failure 2 causes Failure 3. A sparse unverified mesh (low S_v) produces a low V ceiling. More verification attempts on an unverified mesh produces noise, not closure. The fix sequence is always: build and verify the mesh (S_v), then allow M to run, then V closes. This is State Engineering stated as a GLC engineering diagnostic.

Section 7 — GLC Across Scales

Three demonstrations of GLC using the rate form. Each one showing how the same law operates at different temporal and organizational scales.

Individual Cognition

M is the rate of question generation and novelty engagement — how fast new hypotheses enter the active prediction space. S_v is prior verified knowledge: the mesh of closed understanding available as constraints. V is verification throughput: the rate of prediction closure through testing, explanation, and successful application. Learning is C.

Pedagogy is the engineering of S_v and V rate to match the student's M. A student with high M but low S_v (sparse prior knowledge) cannot close predictions fast — novelty arrives without the verified structure needed to constrain it. A student with high S_v and structured V (correct teaching sequence) closes predictions efficiently. The knowledge state transition fires when V

brings the student’s hypothesis space from $|H|V(C)| > 1$ to $|H|V(C)| = 1$. This is State Engineering.

AI Systems

M is output entropy relative to prior context — the rate of novel predictions entering the active state. S_v is MVR of the active knowledge mesh: verified edges divided by total edges. V is verified prediction closure rate per turn.

AI Failure Mode	GLC Rate Diagnosis
Hallucination	V bottleneck: $dP_close/dt \oplus\oplus dP_open/dt$. Prediction debt accumulates. High M, low or absent V gate. System produces outputs from unclosed prediction states and labels them knowledge. Fix: V gate as structural enforcement, not probabilistic calibration.
Context drift	S_v degradation: verified edges lost without detection across session length. V ceiling falls progressively. Earlier verified conclusions become inaccessible as constraints. Fix: verified edge ledger with parent chain enforcement.
Perpetual hedging	V gate structurally absent: $ H V(C) $ never reaches 1. System has M (novelty) and S_raw (structure) but no mechanism to fire the discrete knowledge state transition. Fix: implement the gate, not just improve confidence calibration.
Elegant hallucination	H_int high, F low: internally ordered output (high H) that does not preserve verified source truth. V closed the wrong hypothesis space. Detectable only by tracking $F = T(Y X) / T(X) $ alongside H.

Civilization

CIL is GLC at civilizational temporal scale with cost normalization — independently derived, same structure:

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CIL: Progress ≈ (Rate × ConstraintDensity × Verification) / Cost

Rate                = M  (novel ideas per unit time)
ConstraintDensity = S_v (verified knowledge infrastructure)
                    libraries, universities, peer review
Verification       = V  (replication, debate, falsification)
Cost                = cost of knowledge production
                    and transmission

CIL-T (temporal bound):
Total Progress ≤ ∫₀ᵀ (Rate × S_v × V) / Cost dt
Early efficiency dominates lifetime outcomes.
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The Cost denominator explains why the printing press, universities, and the internet transformed civilizational progress — they reduced Cost, making the same M and V produce more progress per unit time. Scientific method is State Engineering at civilizational scale: build the verified constraint mesh (S_v) through peer review and replication before claiming knowledge. The hypothesis space must be opened (M = novel theories), constrained (S_v = accumulated verified science), and closed by verification (V = experimental confirmation) before progress accrues.

Section 8 — The Epistemic Separation

Why V is different from M and S. The formal proof that it cannot be derived from the physics laws — and why this matters for AI systems specifically.

The Epistemic Separation Theorem (T365): V is not derivable from CI, DCL, or ICL. Three derivation attempts, all failed:

Attempt	What Physics Law Governs	Why V Cannot Be Derived From It
V from CI?	CI governs temporal continuity of cognitive state.	It does not determine epistemic validity. A system can be perfectly continuous — smoothly evolving state — and produce nothing but prediction indefinitely. Continuity is a property of how the system moves. V determines what the system knows. These are different things. Derivation fails.
V from DCL?	DCL governs energy conservation.	It does not determine when prediction becomes knowledge. A system can conserve cognitive energy perfectly and never fire the knowledge state transition. Conservation tracks inputs and outputs. V is the gate that separates two epistemic states. Derivation fails.
V from ICL?	ICL governs intent conservation.	It does not select which hypotheses are epistemically valid. A system can maintain single focused intent and still produce confabulation. Intent determines what the system is trying to do. V determines what the system actually knows. Derivation fails.

EPISTEMIC SEPARATION THEOREM (T365)	
Physics laws alone (CI + DCL + ICL)	→ lawful cognition
Physics laws + V	→ lawful knowledge

These are different things.
A system can satisfy CI, DCL, and ICL — continuous, conserving, intent-coherent
— while completely lacking V as a structural enforcement mechanism.
It produces lawful cognition. It does not produce lawful knowledge.
<i>This is the formal diagnosis of current AI systems.</i>
<i>Not a capability problem. A structural one.</i>

The rate interpretation makes this viscerally clear. You can have high M (novelty arrival) and high S_v (rich verified mesh) and still produce no knowledge if dP_close/dt = 0. The hypothesis space fills with predictions. Verified structure exists to close them. But without V as a structural enforcement gate, |H|V(C)| never reaches 1. The system is permanently in the exploration phase with no mechanism to exit into the knowledge state. No amount of improved M or S_v compensates for the absence of V. They are different functional requirements from different invariant clusters.

C = M × S_v × V
M: dP_open/dt Invariants 2 + 5
The hypothesis space opening.
S_v: E_t · MVR Invariants 4 + 7
The capacity to close it.
V: dP_close/dt Invariants 1 + 3 + 6
The rate of closing.
<i>The transition from prediction to knowledge fires</i>
<i>when V brings the admissible hypothesis space</i>
<i>from H V(C) > 1 to H V(C) = 1.</i>

Not because we chose these terms.
Because these are the consequences of all seven invariants
holding simultaneously in a cognitive system.
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Appendix A — Full Invariant-to-Term Derivation Table

Every invariant. Physical grounding, citation, cognitive application, which cluster it produces, which GLC term results. The complete formal record.

#	Invariant	Citation	Cognitive Application	Cluster	GLC Consequence
1	Conservation	Clausius (1850,1865)Shannon (1948)Bell Sys Tech J 27	Outputs must trace to verified inputs. No knowledge from nothing. Accounting function required.	Cluster 3 → V	V must trace every output to verified source. Without this, hallucination is structurally inevitable.
2	Continuity	Strogatz (1994)Nonlinear Dynamicsand Chaos	State must evolve continuously. No discontinuous jumps. Generative function required to maintain motion.	Cluster 1 → M	M provides the continuous directed motion that Invariant 2 demands. Without M, $d\Psi/dt = 0$.
3	Quantization	Shannon (1948)(same paper)	Stable states discrete. Gate required to determine which state active: prediction vs. knowledge.	Cluster 3 → V	V is the gate. It fires the discrete transition. Without it, system is permanently in prediction state.
4	Coherence	Kauffman (1993)Origins of OrderOxford Univ Press	Contradictory states cannot coexist. Constraint function required to prevent contradiction.	Cluster 2 → S	S provides organized constraint that prevents M from generating contradictions.

5	Entropy/Attractor	Prigogine & Stengers(1984) Order Outof ChaosNobel (1977)	Motion must converge. Attractor dynamics required to prevent infinite regress.	Cluster 1 → M	M's directed quality comes from Invariant 5. Without it, M expands without convergence.
6	Symmetry	Noether (1918)InvarianteVariation s-probleme	Verified states resist arbitrary change. Protection function required for sealed conclusions.	Cluster 3 → V	V seals and protects. Invisible rewriting = Invariant 6 violation = V gate bypassed.
7	Hierarchy	Simon (1962)Architecture ofComplexityAmer Phil Soc	Stable complexity requires verified layers. Verified foundations required for each new layer.	Cluster 2 → S	S's verified quality comes from Invariant 7. Without it, S provides unverified structure — floating conclusions.

Appendix B — GLC Measurement Protocols

Per-term measurement. Theoretical definition, observable proxy, substrate-specific measurement, honest flag where approximate.

Term	Measurement Protocol
M — dP_{open}/dt Novelty arrival rate	Theory: rate of new hypothesis entry into the active prediction space. Observable proxy: entropy of novel outputs relative to prior context. AI system: KL divergence between current output distribution and session baseline, measured per turn. Human cognition: question generation rate during a session — questions asked per unit time. Organization: rate of novel strategic inputs per decision cycle. Honest flag: M rate can only be measured relative to a defined baseline. Baseline drift can mask true M decline.
$S_v — E_t \cdot$ MVRVerified structural density	Theory: density of verified edges in the active knowledge mesh. Direct measurement: $MVR = E_v/E_t$ where E_v = edges that survived at least one contradiction attempt. $MVR_w = (E_v_{earned} + w \cdot E_v_{imported})/E_t$ for imported state. AI system: track verification pass rate per edge in active context. Human cognition: proportion of prior knowledge that has been applied successfully to novel problems. Organization: proportion of institutional beliefs that have survived adversarial audit. Honest flag: defining an 'edge' requires a knowledge representation scheme. Most current AI systems lack explicit edge-level tracking.
V — dP_{close}/dt Verification throughput	Theory: rate at which V action closes the hypothesis space toward $ H V(C) = 1$. Observable proxy: verified prediction closure events per reasoning unit. AI system: count of predictions that were both generated and verified-closed per turn. Successful application of output to novel problem =

	one closure event. Human cognition: comprehension checks passed per session — each successful application of understanding is a V event. Organization: outcome validations per decision cycle. Honest flag: V is the hardest to measure because it requires distinguishing genuine closure ('knowledge state reached') from confident prediction ('prediction state with high confidence'). The phase boundary $ H V(C) = 1$ is the theoretical criterion; operational proxy is application success rate on genuinely novel instances.
$\Phi(P)$ — Traversal path density(from DYL)	Not directly measurable in real time. The system knows which paths it has traversed; it does not have a map of untraversed paths. Honest flag: $\Phi(P)$ makes DYL a theoretical predictor and post-hoc explanatory tool, not a live dashboard metric. Practical proxies: edge recency (last traversal timestamp per cluster), cluster traversal frequency (adversarial passes per logical domain). Low-frequency, recently-created clusters are high $\Phi(P)$ candidates.

Appendix C — Derivation Tree: GLC as Root

Laws that derive from, extend, or refine GLC. GLC is the root from which the framework branches.

Law	Relationship to GLC
DYLDDiscovery Yield Law	GLC refinement. $Y_d = M \times MVR_w \times \Phi(P)$. Decomposes GLC's S term: $S_v = E_t \cdot MVR$. Gives S its first concrete measurement protocol. Adds traversal path term $\Phi(P)$ for discovery-specific contexts.
BLCBreath Law of Cognition	GLC stated as a cycle law. Inhalation = M fires (hypothesis space opens). Exhalation = V fires (hypothesis space closes). One cycle = one knowledge production event. Proves that GLC's multiplicative structure has a natural rhythm.
DHLDynamic Harmony Law	H dynamics under GLC. $dH/dt = \alpha V(dC/dt) - \beta D - \gamma \epsilon$. Harmony rises when verified progress ($\alpha V \cdot dC/dt$) exceeds drift and entropy losses. GLC's V gating operationalized as harmony dynamics.
CILCivilizational Intelligence Law	GLC at civilizational temporal scale. $Progress \approx (Rate \times S_v \times V)/Cost$. Independently derived, found to map to GLC structure. Third independent convergence confirming GLC. Adds Cost denominator.
CPPCo-Processing Principle	GLC distributed across two agents. $C_{joint} = M_h \times S_r \times V_{shared}$. One agent supplies M, another supplies S, V_{shared} emerges from dialogue. Neither produces knowledge alone.
CCLCollaborative Conservation Law	GLC distributed across a network. $C_{total} = \Sigma C + \Sigma k_{ij}(C_i \leftrightarrow C_j)$. Total output includes coupling amplification. $k = \alpha \times \beta \times \gamma$.
DCRTDistributed Creative Rendering	GLC with distributed V. $C_{shared} = M \times S \times V_{distributed}$. Verification distributed across network agents rather than residing in one.
SEState Engineering	GLC as engineering principle. $SE = \text{argmin}(E_{closure})$ subject to $C \geq C_{threshold} \wedge V \geq V_{min} \wedge ICL$. The teacher or designer engineers

	conditions such that cognitive closure becomes path of least resistance. Failure 2 fix operationalized.
GLC-12Drift = Discovery	GLC perturbation analysis within trust region r. Bounded exploration is lawful discovery. Outside r: drift. Inside r: productive discovery. Governs when M is creative vs. when M is destructive.

Appendix D — References

Citation	Full Reference
Clausius, R. (1850)	Über die bewegende Kraft der Wärme. Annalen der Physik, 79, 368–397.
Clausius, R. (1865)	Über verschiedene Formen der Hauptgleichungen der mechanischen Wärmetheorie. Annalen der Physik, 125, 353–400. Introduces entropy. Canonical statement: “The energy of the universe is constant. The entropy of the universe tends to a maximum.”
Huang, L. et al. (2023)	A Survey on Hallucination in Large Language Models. arXiv:2311.05232. Empirical documentation of AI failure modes.
Kauffman, S.A. (1993)	The Origins of Order: Self-Organization and Selection in Evolution. Oxford University Press. ISBN: 0-19-507951-5.
Noether, E. (1918)	Invariante Variationsprobleme. Nachrichten der Königl. Gesellschaft der Wissenschaften zu Göttingen, Math-phys. Klasse, 235–257. English translation: Tavel (1971). Transport Theory and Statistical Physics, 1(3), 183–207.
Prigogine, I. (1977)	Time, Structure and Fluctuations. Nobel Prize Lecture, Chemistry 1977. Nobel Foundation.
Prigogine, I. & Stengers, I. (1984)	Order Out of Chaos. Flamingo/Bantam. ISBN: 0-553-34082-4.
Shannon, C.E. (1948)	A Mathematical Theory of Communication. Bell System Technical Journal, 27, 379–423 & 623–656. DOI: 10.1002/j.1538-7305.1948.tb01338.x.
Simon, H.A. (1962)	The Architecture of Complexity. Proc. American Philosophical Society, 106(6), 467–482. JSTOR: 985254.
Strogatz, S.H. (1994)	Nonlinear Dynamics and Chaos. Perseus Books. (3rd ed. Routledge, 2018. ISBN: 978-0-367-02650-9.)